

Marcelo E. Andia K.

CONTACT INFORMATION

Radiology Department & Biomedical Imaging Centre
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EDUCATION

Ph.D.
 Division of Imaging Sciences & Biomedical Engineering, School of Medicine
 King's College London, University of London, UK, 2012

M.D.
 School of Medicine, Pontificia Universidad Católica de Chile, 2006

M.Sc.
 School of Engineering, Pontificia Universidad Católica de Chile, 1996

B.Sc. Mechanical Engineer
 School of Engineering, Pontificia Universidad Católica de Chile, 1994

ACADEMIC EXPERIENCE

2021 – present. Director
 Biomedical Imaging Centre. Pontificia Universidad Católica de Chile.

2017 – present. Associate Professor.
 Radiology Department, School of Medicine. Pontificia Universidad Católica de Chile.

2007 – 2016. Assistant Professor.
 Radiology Department, School of Medicine. Pontificia Universidad Católica de Chile.

1996 – 2005. Instructor Professor.
 Mining Centre, School of Engineering. Pontificia Universidad Católica de Chile.

RESEARCH INTERESTS

Magnetic Resonance Imaging & Spectroscopy, Molecular Imaging, Atherosclerosis, Inflammation.

SOCIETIES

- International Society for Magnetic Resonance in Medicine (ISMRM)*
- Society for Cardiovascular Magnetic Resonance (SCMR)*
- European Society for Magnetic Resonance in Medicine and Biology (ESMRMB)*

RESEARCH GRANTS

1. 2026-2030. *Principal Investigator*, Metabolic ageing: phenotyping healthy ageing using quantitative magnetic resonance imaging and Machine Learning. *Fondecyt Regular 1260578 (US250K)*
2. 2026-2028. *Principal Investigator*, Robust and interpretable data-driven learning of dynamical systems for the discovery of regulation mechanisms for biological signals. *Avanza-UC AV-25053 (US30K)*.
3. 2025-2029. *Co-Investigator*, Liver transplant allocation: integrating muscle health into decision-making. *Fondecyt Exploracion 13250065 (US400K)*, P.I.: Dr. Daniel Cabrera, School of Medicine, Universidad de Los Andes.
4. 2025-2027. *Co-Investigator*, Preclinical evaluation of Neurotinib's protective effects on seizures and secondary brain damage in epilepsy mouse models induced by kainic acid, traumatic brain injury, and viral encephalitis. *Fondef IDeA 25i10566 (US250K)*, P.I.: Dr. Alejandra Alvarez, Faculty of Biological Sciences, PUC.
5. 2025-2027. *Co-Investigator*, Evaluación y escalamiento de un modelo de gestión clínica basado en el uso de los grupos relacionados por el diagnóstico (GRD) para la mejora de la eficiencia de la producción de hospitales de alta complejidad en Chile. *Fondef IDeA 25i10300 (US250K)*, P.I.: Dr. Fernando Poblete, Public Health Department, PUC.
6. 2024-2028. *Deputy Director*, Sound Renaissance in Medicine: Recovering the Power of Sound Waves for Cardiopulmonary Disease Monitoring and Autonomous Intelligent Ultrasound. *Fondecyt Exploracion 13240059 (US400K)*
7. 2022-2026. *Principal Investigator*, Improving metabolic phenotyping of MAFLD patients using Magnetic Resonance Spectroscopy and Machine Learning. *Fondecyt Regular 1220922 (US300K)*
8. 2022-2023. *Co-Investigator*, Barreras y facilitadores para la educación en Inteligencia Artificial en la Carrera de Medicina. *Fondedoc-UC (US6K)*. P.I.: Dr. Pablo Muñoz, Radio-oncology Department, PUC.
9. 2021-2031. *Deputy Director*, Millennium Institute for Intelligent Healthcare Engineering (iHEALTH) (US12M). P.I.: Dr. Claudia Prieto, Institute for Mathematical and Computational Engineering, PUC.
10. 2019-2023. *Co-Investigator*, Novel radiologic and metabolomic profiling of adrenal tumors to identify cardiometabolic risk. *Fondecyt Regular 1190419 (US300K)*. P.I.: Dr. René Baudrand, Endocrinology Department, PUC.
11. 2019-2022. *Co-Investigator*, Development and evaluation of a clinical management tool for the improvement of the quality and efficiency of the production of large hospitals in Chile, based on diagnostic related groups (DRG). *Fondef IDeA 19i10077 (US250K)*, P.I.: Dr. Fernando Poblete, Public Health Department, PUC.
12. 2019-2022. *Co-Investigator*, Multimodal assessment of acute cardiac toxicity induced by thoracic radiotherapy in cancer patients. *Fondecyt Inicio 11190071 (US140K)*. P.I.: Dr. Tomás Merino, Oncology Department, PUC.
13. 2018-2022. *Deputy Director*, Millennium Nucleus in Cardiovascular Magnetic Resonance (US800K). P.I.: Dr. Sergio Uribe, Radiology Department, PUC.
14. 2018-2022. *Principal Investigator*, Understanding the relationship between venous thrombosis and atherosclerosis: the monocyte link. *Fondecyt Regular 1180525 (US300K)*
15. 2018-2022. *Co-Investigator*, Cardiovascular MR function: new insights from intracardiac flows and

- strain analysis. *Fondecyt Regular 1181057 (US300K)*. P.I.: Dr. Sergio Uribe, Radiology Department, PUC.
16. 2018-2022. *Co-Investigator*, Time-course modelling of the vascular epigenetic programming by fetal growth restriction: role of endothelial-derived miRNAs on hypoxia- and oxidative stress-induced responses. *Fondecyt Regular 1181341 (US300K)*. P.I.: Dr. Bernardo Krause, Paediatrics Department, PUC.
 17. 2017-2021. *Co-Investigator*, Studies on the contribution of endothelial dysfunction and platelet activation to the pathogenesis of thrombosis in inflammatory bowel disease. *Fondecyt Regular 1171637 (US280K)*. P.I.: Dr. Jaime Pereira, Haematology Department, PUC.
 18. 2016-2019. *International Collaborator*, Multi-sequence MRI characterization of deep vein thrombosis in man. *British Heart Foundation, BHF15/89/31793 (US200K)*, P.I. Dr. Alkystis Phinikaridou, Division of Imaging Sciences, King's College London, UK.
 19. 2016-2019. *Co-Investigator*, Sonification of medical data. *Fondecyt Regular 1161328 (US240K)*, P.I.: Dr. Rodrigo Cádiz, Music Institute and Electric Engineering Department, PUC.
 20. 2016-2019. *Co-Investigator*, Free breathing 3D Cardiac Multiparametric MR Fingerprinting. *Fondecyt Regular 1161055 (US220K)*, P.I.: Dr. Claudia Prieto, Electrical Engineering Department, PUC.
 21. 2016-2018. *Co-Investigator*, Improving Quantitative Susceptibility Mapping reconstruction methods based on MRI acquisitions. *Fondecyt Regular 1161448 (US180K)*, P.I.: Dr. Cristian Tejos, Electrical Engineering Department, PUC.
 22. 2015-2018. *Associate Researcher*, Magnetic resonance imaging technology for aging related diseases: brain, heart, and vessels. *Anillo ACT1416 (US650K)*, P.I.: Dr. Pablo Irrázaval, Electrical Engineering Department, PUC.
 23. 2014-2017. *Co-Investigator*, Non-invasive 3D full-field quantification of cardiovascular 4D flow MRI. *Fondecyt 1141036 (US240K)*, P.I.: Dr. Sergio Uribe, Radiology Department, PUC.
 24. 2014-2016. *Co-Investigator*, Studies on the mechanisms involved in the pathogenesis of cocaine-associated brain perfusion defects: the role of endothelial dysfunction and activation of haemostatic system. *Fondecyt 1141051 (US280K)*, P.I.: Dr. Jaime Pereira, Haematology Department, PUC.
 25. 2014-2016. *Co-Investigator*, Design of a laboratory non-invasive, non-destructive procedure for copper sulphide heap leaching process optimization. *Fondecyt 11130540 (US140K)*, P.I.: Dr. Alvaro Videla, Mining Engineering Department, PUC.
 26. 2013-2016. *Principal Investigator*, Development of contrast-free magnetic resonance angiography and tissue perfusion techniques. *Fondecyt Regular 1130379 (US180K)*
 27. 2013-2015. *Co-Investigator*, Design of a health communication system for the elderly and their support network. *Fondef IDeA 13i10210 (US240K)*, P.I.: Dr. Sergio Godoy, Communication Faculty, PUC.
 28. 2013-2015. *Co-Investigator*, Topologically flexible prior shape knowledge for level set segmentations. *Fondecyt Regular 1130887 (US140K)*, P.I.: Dr. Cristian Tejos, Electrical Engineering Department, PUC.
 29. 2013-2015. *Co-Investigator*, f-MRI plus EEG implementation for the study of brain conditioning throws brain-computer interface. *Interdisciplinary VRI Grant. Vicerrectoría de Investigación, Pontificia Universidad Católica de Chile (US20K)*, P.I.: Dr. Cristian Tejos, Electrical Engineering Department, PUC.
 30. 2012-2013. *Principal Investigator*, International network for research on cardiovascular magnetic resonance, King's College London – Biomedical Imaging Centre UC. *Redes120026 (US35K)*
 31. 2010-2013. *Associate Researcher*, Biomedical imaging in obesity: from cellular mechanisms to clinical

applications. *Anillo ACT79 (US800K)*, P.I.: Dr. Pablo Irarrázaval, Electrical Engineering Department, PUC.

32. 2009–2010. Co-Investigator, Evaluation of the stereotaxic radiosurgery effects using MRI. *Pilot Grant in General Aspects of Magnetic Resonance in Medicine ISMRM (US25K)*, P.I.: Dr. Cristian Tejos, Electrical Engineering Department, PUC.

PUBLICATIONS

1. **Andia ME**, Garrido F, Sanchez I. Impact and Future Outlook of Artificial Intelligence in Universities and Medical Education. *Rev Med Chile* 2026;
2. Keeling G, Wang X, Chen W, Chaher N, Gao L, **Andia ME**, Tumanov S, Golda P, Chowdhury MM, Saha P, Livieratos L, Nadel J, Stocker R, Phinikaridou A. Development and validation of an activatable PET radiotracer reporting extracellular myeloperoxidase activity for the detection of unstable atherosclerotic plaque. *npj Imaging* 2026;4(1):23
3. Castillo-Triana N, Juri C, Gonzalez-Mendez P, Cadiz RF, **Andia ME**. Neural substrates of speech disorders in Parkinson's disease: a narrative review of the evidence from task-based and resting-state fMRI studies. *Brain Lang* 2026; 275:e105720
4. Munoz N, Garcia M, Cuiza A, Tepper A, Vasquez J, Ramirez J, Barbagelata D, Aguirre J, Maldonado ME, Ornstein C, Fritsch R, Repetto G, Milovic C, **Andia ME**, Crossley N, Tejos C. Quantitative susceptibility mapping of deep brain nuclei in 22q11.2 deletion syndrome. *Front Psychiatry* 2026; 16:e1652700
5. Silickas J, Smith A, **Andia ME**, Botnar R, Modarai B, Karunanithy N, Patel A, Black S, Saha P, Phinikaridou A. Multi-parametric contrast-free MRI successfully identifies venous thrombus responsive to lytic therapy: from mice to humans. *Circ Cardiovasc Imaging* 2025; 18(11):e018175
6. Jimenez-Jara C, Salas R, Diaz-Navarro R, Chabert S, **Andia ME**, Vega J, Urbina J, Uribe S, Sekine T, Raimondi F, Sotelo J. AI applied to Cardiac Magnetic Resonance for Precision Medicine in Coronary Artery Disease: A Systematic Review. *J Cardiovasc Dev Dis* 2025; 12(9):345
7. Zamora T, Salas P, Zuñiga S, Botello E, **Andia ME**. Generative Artificial Intelligence, Large Language Models and ChatGPT in Musculoskeletal Oncology: Current Applications and Future Potential. *J Clin Orthop Trauma* 2025; 69:103161
8. Alfaro-Nasta E, Gonzalez-Mendez P, Lorenzoni J, Juri C, **Andia ME**. Predictive Value of Brain Volumetry in the Response to Subthalamotomy using High-Intensity Focused Ultrasound Treatment in Patients with Parkinson's Disease. *Parkinsons Dis* 2025: 8780938
9. Besa P, Alegria A, Gonzalez N, Biancardi F, Vidal C, Cikutovic P, **Andia ME**, Mura J. Regional Bone Density Patterns of the Tibial Plateau: Implications for Finite Element Analysis. *Front Bioeng Biotechnol* 2025;13:1541536.
10. Rivera K, González L, Parra L, Oyarzún JE, Concepción-Alvarez A, Costa de Camargo A, Bridi R, Rigotti A, **Andia ME**. Red Wine Grape Pomace Modulates Gut Barrier Integrity and Enhances Survival in Diet-Induced Ischemic Heart Disease: Role of Insoluble-Bound Phenolics. *Antioxidants* 2025; 14(5):574
11. Manjarres L, Xavier A, Gonzalez L, Garrido C, Zacconi FC, Rivera K, Parra L, Phinikaridou A, Besa C, **Andia ME**. Sex differences in the relationship between body composition and MASLD progression in a murine model of metabolic syndrome. *iScience* 2025; 28(2):111863
12. Xavier A, Oyarzun JE, Zacconi F, Zanolungo S, **Andia ME**. Liver Magnetic Resonance Spectroscopy as

- an alternative for evaluating Niemann-Pick C disease progression. *RSC Adv* 2025; 15(6):4079-85
13. Meneses JP, Qadir A, Surendran N, Arrieta C, Tejos C, **Andia ME**, Chen Z, Uribe S. Unbiased and reproducible liver MRI-PDFF estimation using a scan protocol-informed deep learning method. *Eur Radiol* 2025; 35(5):2843-2854
 14. Rivera K, Gonzalez L, Bravo L, Manjarrés L, **Andia ME**. The Gut-Heart Axis: Molecular Perspectives and Implications for Myocardial Infarction. *Int J Mol Sci* 2024; 25(22):12465
 15. Núñez-Pizarro P, Montenegro G, Núñez G, **Andia ME**, Espinosa-Bustos C, Costa de Camargo A, Oyarzún JE, Bridi R. Comparative Study of Phenolic Content, Antioxidant, and Hepatoprotective Activities of Unifloral Quillay Tree (*Quillaja saponaria* Molina) and Multifloral Honeys from Chile. *Plants* 2024;13(2):3187
 16. Echenique MB, Godoy E, Cadiz R, **Andia ME**. Enhancement of cardiac and respiratory sounds for cell phone reproduction by means of digital sound processing methods. *Pers Ubiquitous Computer* 2024;28(5):845-856
 17. Ramos-Zaldívar H, Polakovicova I, Salas-Huenuleo E, Yefi CP, Silva D, Jara-Guajardo P, Oyarzun JE, Neira-Troncoso A, Burgos PV, Cavieres V, Arias-Munoz E, Martinez C, Riveros AL, Corvalán AH, Kogan MJ, **Andia ME**. The cervical and meningeal lymphatic network as a pathway for retrograde nanoparticle transport to the brain. *Int J Nanomedicine* 2024;19:10725–10743
 18. Saha P, Gutmann C, Kingdon J, Dregan A, Bertolaccini L, Grover SP, Patel AS, Modarai B, Lyons O, Schulz C, **Andia ME**, Phinikaridou A, Botnar RM, Smith A. Venous thrombosis accelerates atherosclerosis in mice. *Circulation* 2023;147(25):1945-1947
 19. Bravo L, Pereyra KV, Diaz HS, Flores M, Schwarz KG, Toledo C, Díaz-Jara E, González L, **Andia ME**, Del Rio R. Enhanced peripheral chemoreflex drive is associated with cardiorespiratory disorders in mice with coronary heart disease. *Adv Exp Med Biol* 2023;1427:99-106.
 20. Lavin-Plaza B, Eykyn T, Phinikaridou A, Xavier A, Kumar S, Buqué X, Aspichueta P, Sing-Long C, Arrese M, Botnar RM, **Andia ME**. Characterization of hepatic fatty acids using Magnetic Resonance Spectroscopy for the assessment of treatment response to Metformin in an eNOS^{-/-} mouse model of metabolic NAFLD-NASH. *NMR Biomed* 2023;36(8):e4932.
 21. Gonzalez L, Rivera K, **Andia ME**, Martinez G. The IL-1 family and its role in atherosclerosis. *Int J Mol Sci* 2023; 24, 17.
 22. Messina P, Pino P, Parra D, Soto A, Besa C, Uribe S, **Andia ME**, Tejos C, Prieto C, Capurro D. A survey on deep learning and explainability for automatic report generation from medical images. *ACM Computing Surveys (CSUR)* 2022; 54, 10s, 1–40.
 23. Ramos-Zaldivar H, Reyes-Perdomo K, Espinoza-Moreno N, Dox-Cruz E,..., **Andia ME**. Safety and efficacy of thymic peptides in the treatment of hospitalized Covid-19 patients in Honduras. *Georgian Med News* 2022;(330):99-105.
 24. Ramos-Zaldivar HM, Polakovicova I, Salas-Huenuleo E, Corvalán AH, Kogan MJ, Yefi CP, **Andia ME**. Extracellular vesicles through the blood-brain barrier: a review. *Fluids Barriers CNS* 2022; 19(1):60.
 25. Acevedo F, Ip T, Orellana M, Martinez G, Gabrieli L, **Andia ME**, Besa C, Pinto MP, Sanchez C, Merino T. Oncological Benefit versus Cardiovascular Risk in Breast Cancer Patients Treated with Modern Radiotherapy. *J Clin Med* 2022;11(13):3889.

26. Costa de Camargo A, Alvarez A, Arias-Santé MF, Oyarzún JE, **Andia ME**, Uribe S, Núñez Pizarro P, Bustos SM, Schwember AR, Shahidi F, Bridi R. Soluble Free, Esterified and Insoluble-Bound Phenolic Antioxidants from Chickpeas Prevent Cytotoxicity in Human Hepatoma HuH-7 Cells Induced by Peroxyl Radicals. *Antioxidants* 2022;11(6):e1139
27. Contreras-Briceño F, Espinosa-Ramirez M, Keim-Bagnara V, Carreño-Román M, Rodríguez-Villagra R, Villegas-Belmar F, Viscor G, Gabrielli L, **Andia ME**, Araneda OF, Hurtado DE. Determination of the Respiratory Compensation Point by Detecting Changes in Intercostal Muscles Oxygenation by Using Near-Infrared Spectroscopy. *Life* 2022;12(3):444
28. Denecken E, Sotelo J, Arrieta C, **Andia ME**, Uribe S. Impact of Respiratory Gating on Hemodynamic Parameters from 4D Flow MRI. *Appl Sci* 2022;12, 2943.
29. Lavin B, **Andia ME**, Saha P, Botnar R, Phinikaridou A. Quantitative MRI of Endothelial Permeability and (Dys)function in Atherosclerosis. *J Vis Exp* 2021; Dec(178).
30. Merino T, Pinto MP, Orellana MP, Martinez G, **Andia ME**, Munoz-Schuffenegger P, Acevedo F, Gabrielli L, Sanchez C, Pereira J. Multimodal assessment of acute cardiac toxicity induced by thoracic radiotherapy in cancer patients. Study protocol. *BMC Cancer* 2021;21(1):1114.
31. Díaz-Jara E, Díaz HS, Rios A, Ortolani D, Andrade DC, Toledo C, Pereyra KV, Schwarz K, Ramirez G, Arias P, Ortiz FC, **Andia ME**, Del Rio R. Exercise training reduces brainstem oxidative stress and restores normal breathing function in heart failure. *Free Radic Biol Med* 2021;172:470-481
32. Xavier A, Zacconi F, Santana-Romo F, Eykyn TR, Lavin B, Phinikaridou A, Botnar R, Uribe S, Oyarzún JE, Cabrera D, Arrese M, **Andia ME**. Assessment of hepatic fatty acids during non-alcoholic steatohepatitis progression using magnetic resonance spectroscopy. *Ann Hepatol* 2021;25:100358
33. Rivera K, Quiñones V, Amigo L, Santander M, Salas-Pérez F, Xavier A, Fernández-Galilea M, Carrasco G, Cabrera D, Arrese M, Busso D, **Andia ME**, Rigotti A. Lipoprotein receptor SR-B1 deficiency enhances adipose tissue inflammation and reduces susceptibility to hepatic steatosis during diet-induced obesity in mice. *Biochim Biophys Acta Mol Cell Biol Lipids* 2021; 1866(6):158909
34. Arrieta C, Sing-Long CA, Mura J, Irrarrázaval P, **Andia ME**, Uribe S, Tejos C. Level set segmentation with shape prior knowledge using intrinsic rotation, translation and scaling alignment. *Biomed Signal Process Control* 2021;63:e102241
35. Oyarzun JE, **Andia ME**, Uribe S, Nuñez-Pizarro P, Nuñez G, Montenegro G, Bridi R. Honeybee pollen extracts reduce oxidative stress and steatosis in hepatic cells. *Molecules* 2021; 26(1):e26010006
36. Bridi R, von Poser GL, Gomez M, **Andia ME**, Oyarzún JE, Nuñez P, Vasquez-Arias AJ, Espinosa-Bustos C. Hepatoprotective species from the Chilean medicinal flora: *Junellia spathulata* (Verbenaceae). *J Ethnopharmacol* 2021; 267:113543
37. Lavin B, Phinikaridou A, **Andia ME**, Potter M, Lorrio S, Rashid I, Botnar RM. Sustained focal vascular inflammation accelerates atherosclerosis in remote arteries. *Arterioscler Thromb Vasc Biol* 2020; 40(9):2159-2170.
38. Xavier A, Arteaga de Castro C, **Andia ME**, Luijten PR, Klomp DW, Fillmer A, Prompers JJ. Metabolite cycled liver ¹H MRS on a 7T parallel transmit system. *NMR Biomed* 2020; 33(8):e4343.
39. Arrieta C, Urrutia J, Besa P, Montalba C, Lafont N, **Andia ME**, Uribe S. Automatic quantification of fat infiltration in paraspinal muscles using T2-weighted images: An Osirix application. *Biomed Signal Process Control* 2020; 57:e101793.

40. Lavin B, Lacerda S; **Andia ME**, Lorrio S, Bakewell R, Smith A, Rashid I, Botnar RM, Phinikaridou A. Tropoelastin: An in vivo imaging marker of dysfunctional matrix turnover during abdominal aortic dilatation. *Cardiovasc Res* 2020;116(5):995-1005.
41. Xavier A, Zacconi F, Gainza C, Cabrera D, Arrese M, Uribe S, Sing-Long C, **Andia ME**. Intrahepatic fatty acids composition as a biomarker of NAFLD progression from steatosis to NASH by using ¹H-MRS. *RSC Adv* 2019; 9(72):42132-42139.
42. Rivera K, Salas-Pérez F, Echeverría G, Urquiaga I, Dicenta S, Pérez D, de la Cerda P, González L, **Andia ME**, Uribe S, Tejos C, Martínez G, Busso D, Irrazaval P, Rigotti A. Red wine grape pomace attenuates atherosclerosis and myocardial damage and increases survival in association with improved plasma antioxidant activity in a murine model of lethal ischemic heart disease. *Nutrients* 2019; 11(9):e002135.
43. Ramos IT, Henningsson M, Nezafat M, Lavin B, Lorrio S, Gebhardt P, Protti A, Eykyn TR, **Andia ME**, Flögel U, Phinikaridou A, Shah AM, Botnar RM. Simultaneous assessment of cardiac inflammation and extracellular matrix remodelling after myocardial infarction. *Circ Cardiovasc Imaging* 2018; 11(11):e007453.
44. Contreras-Duarte S, Chen P, **Andia ME**, Uribe S, Irrazaval P, Kopp S, Kern S, Marsche G, Busso D, Wadsack C, Rigotti A. Attenuation of atherogenic apo B-48-dependent hyperlipidemia and high density lipoprotein remodeling induced by vitamin C and E combination and their beneficial effect on lethal ischemic heart disease in mice. *Biol Res* 2018;15(1):34-44.
45. Campos F, Abrigo J, Aguirre F, Garcés B, Arrese M, Karpen S, Cabrera D, **Andia ME**, Simon F, Cabello-Verrugio C. Sarcopenia in a mice model of chronic liver disease: Role of the ubiquitin-proteasome system and oxidative stress. *Pflugers Arch - Eur J Physiol* 2018; 470(10):1503-1519.
46. Phinikaridou A, Lacerda S, Lavin B, **Andia ME**, Smith A, Saha P, Botnar RM. Tropoelastin: A novel marker for plaque progression and instability. *Circ Cardiovasc Imaging* 2018; 11(8):e007303.
47. Urrutia J, Besa P, Lobos D, **Andia ME**, Arrieta C, Uribe S. Is a single level measurement of paraspinal muscle fat infiltration and cross-sectional area representative of the entire lumbar spine?. *Skeletal Radiology* 2018;47(7):939-945.
48. Urrutia J, Besa P, Lobos D, Campos M, Arrieta C, **Andia ME**, Uribe S. Lumbar paraspinal muscle fat infiltration is independently associated with sex, age, and inter-vertebral disc degeneration in symptomatic patients. *Skeletal Radiology* 2018;47(7):955-961.
49. Milovic C, Acosta-Cabronero J, Pinto JM, Mattern H, **Andia ME**, Uribe S, Tejos C. A new discrete dipole kernel for quantitative susceptibility mapping. *Magn Reson Imaging* 2018;51:7-13.
50. Navarro ME, Razmilic D, Araos I, Rodrigo A, **Andia ME**. Contrast-enhanced spectral mammography: experience in 465 examinations. *Rev Med Chile* 2018;146(2):141-149.
51. Montalba C, Urbina J, Sotelo J, **Andia ME**, Tejos C, Irrazaval P, Valverde I, Uribe S. Variability of 4D flow parameters when subjected to changes in MRI acquisitions parameters using a realistic thoracic aortic phantom. *Magn Reson Med* 2018;79(4):1882-1892.
52. Sotelo J, Urbina J, Valverde I, Mura J, Tejos C, Irrazaval P, **Andia ME**, Hurtado D, Uribe S. Three-dimensional quantification of vorticity and helicity from 3D cine PC-MRI using finite-element interpolations. *Magn Reson Med* 2018;79(1):541-553.
53. Sotelo J, Bächler P, Urbina J, Crelier G, Toro L, Ferreiro M, Valverde I, **Andia ME**, Tejos C, Irrazaval P, Uribe S. Quantification of pulmonary regurgitation in patients with repaired Tetralogy of Fallot by

- 2D phase-contrast MRI: Differences between the standard method of velocity averaging and a pixel-wise analysis. *JRSM Cardiovasc Dis* 2017;6:e2048004017731986.
54. Zamora T, Palma J, **Andia ME**, Garcia P, Wozniak A, Solar A, Campos M. Effect of *Propionibacterium acnes* (PA) injection on intervertebral disc degeneration in a rat model: does it mimic modic changes?. *Orthop Traumatol Surg Res* 2017;103(5):795-799.
 55. Videla A, Matamala LF, Uribe S, **Andia ME**. Ferrous ion oxidation monitoring by using magnetic resonance imaging for bio-oxidation laboratory testing. *Miner Eng* 2017;106:108-115.
 56. Arrieta C, Uribe S, Sing-Long C, Hurtado D, **Andia ME**, Irarrazaval P, Tejos C. Simultaneous left and right ventricle segmentation using topology preserving level sets. *Biomed Signal Process Control* 2017; 33:88-95.
 57. Phinikaridou A, **Andia ME**, Lavin-Plaza B, Smith A, Saha P, Botnar RM. Increased vascular permeability measured with an albumin-binding MR contrast agent is a surrogate marker of rupture-prone atherosclerotic plaque. *Circ Cardiovasc Imaging* 2016;9(12):e004910.
 58. Mura J, Pino AM, Sotelo J, Valverde I, Tejos C, **Andia ME**, Irarrazaval P, Uribe S. Enhancing the velocity data from 4D flow MR images by reducing its divergence. *IEEE Trans Med Imaging* 2016; 35(10):2353-2364.
 59. Arboleda C, Aguirre-Reyes D, Garcia MP, Tejos C, Muñoz L, Miquel JF, Irarrazaval P, **Andia ME**, Uribe S. Total liver fat quantification using three-dimensional respiratory self-navigated MRI sequence. *Magn Reson Med* 2016;76(5):1400-1409.
 60. Urbina J, Sotelo J, Springmüller D, Montalba C, Letelier K, Tejos C, Irarrazaval P, **Andia ME**, Razavi R, Valverde I, Uribe SA. Realistic aortic phantom to study hemodynamics using MRI and cardiac catheterization in normal and aortic coarctation conditions. *J Magn Reson Imaging* 2016; 44(3):683-697.
 61. Sotelo J, Urbina J, Valverde I, Tejos C, Irarrazaval P, **Andia ME**, Uribe S, Hurtado DE. 3D quantification of wall shear stress and oscillatory shear index using a finite-element method in 3D CINE PC-MRI data of the thoracic aorta. *IEEE Trans Med Imaging* 2016;35(6):1475-1487.
 62. Letelier K, Urbina J, **Andia ME**, Tejos C, Irarrazaval P, Prieto C, Uribe S. Accelerating dual cardiac phase images using undersampled radial phase encoding trajectories. *Magn Reson Imaging* 2016; 34(7):1017-1025.
 63. Aguirre-Reyes DF, Sotelo JA, Arab JP, Arrese M, Tejos R, Irarrazaval P, Tejos C, Uribe SA, **Andia ME**. Intrahepatic portal vein blood volume estimated by non-contrast Magnetic Resonance Imaging for the assessment of portal hypertension. *Magn Reson Imaging* 2015;33(8):970-7.
 64. Passacuale G, Phinikaridou A, Warboys C, Cooper M, Lavin B, Alfieri A, **Andia ME**, Botnar RM, Ferro A. Aspirin-induced histone acetylation in endothelial cells enhances synthesis of the secreted isoform of netrin-1 thus inhibiting monocyte vascular infiltration. *Br J Pharmacol* 2015;172(14):3548-64.
 65. Pinto JM, Arrieta C, **Andia ME**, Uribe S, Ramos-Grez J, Vargas A, Irarrazaval P, Tejos C. Sensitivity analysis of geometric errors in additive manufacturing medical models. *Med Eng Phys* 2015; 37(3):328-34.
 66. Hussain T, Henningsson M, Butzbach B, Lossnitzer D, Greil GF, **Andia ME**, Botnar RM. Combined coronary lumen and vessel wall MRI with i-T2prep: influence of nitroglycerin. *Int J Cardiovasc Imaging* 2015;31(1):77-82.
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82. Andia ME, Saha P, Grover S, Wiethoff AJ, Schaeffter T, Smith A, Botnar RM. Fibrin targeted MRI allows staging of deep venous thrombosis and identifies thrombi amenable for thrombolysis. Proceedings 20th International Conference of the ISMRM, Melbourne, 2012 (Oral Presentation)
83. Andia ME, Henningson M, Hussain T, Phinikaridou A, Greil G, Botnar RM. i-T2prep: Flow-independent 3D whole heart vessel wall imaging using an interleaved T2-preparation acquisition. Proceedings 20th International Conference of the ISMRM, Melbourne, 2012 (Oral Presentation)
84. Andia ME, Botnar RM. Unenhanced MR liver circulation imaging using a Triple Inversion Recovery ASL technique. Proceedings 20th International Conference of the ISMRM, Melbourne, 2012 (Electronic Poster)
85. Hussain T, Henningson M, Butzback B, Andia ME, Botnar RM. Influence of nitroglycerin for combined coronary lumen and vessel wall MRI. Proceedings 20th International Conference of the ISMRM, Melbourne, 2012 (Oral Presentation)
86. Phinikaridou A, Andia ME, Martin De Rosales R, Botnar RM. A novel contrast agent detects tropoelastin fibers in atherosclerotic plaques. Proceedings 20th International Conference of the ISMRM, Melbourne, 2012 (Oral Presentation)
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 108. Lagos GE, Lehuedé JM, Andia ME. Analysis of the Decontamination Plans of the Foundries of: Chuquicamata, Hernán Videla Lira (Paipote), and Ventanas. Proceedings of the IV International Conference on Clean Technologies for the Mining Industry, Santiago, 2000 (Oral Presentation)
 109. Andia ME, Lagos GE. The Challenges Posed by Mine Closure in Chile. Proceedings of the International Conference Copper 99, Phoenix, 1999 (Oral Presentation)
 110. Lagos GE, Andia ME. Benefits of Private Mining Investment in Chile, Proceedings of the International Conference Copper 99, Phoenix, 1999 (Oral Presentation)
 111. Davis M, Andia ME. Tailings Management: Adapting to More Stringent Environmental Standards in Chile. Proceedings of the Tailings and Mine Waste '99 Conference, Boulder, 1999 (Oral Presentation)
 112. Lagos G, Andia ME. Copper Recycling and the Environment. Proceedings of the World Copper Recycling Conference, Brussels, 1997 (Oral Presentation)

CHAPTERS IN BOOKS

1. Lavin-Plaza B, Phinikaridou A, **Andia ME**, Lacerda S, Henningsson M, Makowski M, Botnar RM (2019) Atherosclerotic Plaque Imaging. In: Kwong R., Jerosch-Herold M., Heydari B. (eds) Cardiovascular Magnetic Resonance Imaging. Contemporary Cardiology. Springer, New York, NY. ISBN: 978-1-4939-8839-6
2. Bastias, G, **Andia ME**. (2009) Chapter 2: Plan de Garantías Explícitas en Salud (GES): Revisando la Estrategia, in: Recetas para el Sector Salud, Edited by Castro R, Ediciones LYD, Chile. ISBN: 978-956-7183-33-3
3. Lagos GE, **Andia ME**. (2002) Chapter 7: Minerales e Hidrocarburos, in: Informe País: Estado del Medio Ambiente en Chile. Centro de Análisis de Políticas Públicas, Universidad de Chile. LOM Ediciones. ISBN: 956-282-830-2
4. Wrann A, **Andia ME**, Lagos GE. (2001) Estimación del riesgo de transporte de ácido sulfúrico en Chile. in: Hacia la Integración de Aspectos Ambientales, Económicos y Comerciales del Sector Minero. Ed. Borregaard N, Gana C, CIPMA-IDRC, LOM Ediciones, pp. 93-116. ISBN: 956-706-514-4
5. **Andia ME**, Lagos GE. (2001) Costos de cierre de tranques de relaves en Chile, in: Hacia la Integración de Aspectos Ambientales, Económicos y Comerciales del Sector Minero, 2001 Ed. Borregaard N, Gana C, CIPMA-IDRC, LOM Ediciones, pp. 117-129. ISBN: 956-706-514-4

TEACHING ACTIVITIES

2018 – present. *MDO3031 Basic Biological Processes in Human Diseases*. Seminar professor. School of Medicine, *Master of Research in Health Science*, Pontificia Universidad Católica de Chile

2016 – present. *MCS3030 Thesis Project*. School of Medicine, *Master of Research in Health Science*, Pontificia Universidad Católica de Chile

2015 – present. *IBM1005 Introduction to Biomedical Engineering*. School of Engineering, Pontificia Universidad Católica de Chile

2015 – present. *MED110A Integrated Medical Sciences*. Seminar professor. School of Medicine, Pontificia Universidad Católica de Chile

2013 – present. *FIS119M Physics for Medicine*. Seminar professor. School of Medicine, Pontificia Universidad Católica de Chile

2014 – 2023. *MED810 Scientific Research Methodology*. School of Medicine, Pontificia Universidad Católica de Chile

2013 – 2020. *IMM3620 Mining Efficiency*. Seminar professor. School of Engineering, Pontificia Universidad Católica de Chile

2013 – 2014. *IEE3764 Image Formation*. School of Engineering, Pontificia Universidad Católica de Chile

2012 – present. *Magnetic Resonance Imaging: Seminars for radiology residents*. School of Medicine, Pontificia Universidad Católica de Chile

2007 – 2008. *BIO238M Molecular Biology of the Cell*. Seminar professor. School of Medicine, Pontificia Universidad Católica de Chile

1999 – 2000. *IMM3100 Environmental Management*. School of Engineering, Pontificia Universidad Católica de Chile

1998 – 2004. *IMM2032 Geostatistics*. School of Engineering, Pontificia Universidad Católica de Chile

ACADEMIC POSITIONS

2026 – present. Director of AI for Healthcare. School of Medicine, Pontificia Universidad Católica de Chile

2021 – present. Director, *Biomedical Imaging Centre*. School of Medicine and School of Engineering, Pontificia Universidad Católica de Chile

2022 – 2026. Director of Master and Doctoral Programs. School of Medicine, Pontificia Universidad Católica de Chile

2016 – 2022. Director, *Master of Research in Health Sciences*. School of Medicine, Pontificia Universidad Católica de Chile

2015 – 2022. Coordinator, *Interdisciplinary Undergraduate Research Certificate*. School of Medicine, Pontificia Universidad Católica de Chile

STUDENT SUPERVISION

1. Francisca Urrutia Goldsack (2025– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Supervisor
2. Laura Manjarrés Madrid (2025– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Supervisor
3. Liena Bravo Urquiza (2025– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Supervisor
4. Karol Acevedo Encalada (2024– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Supervisor
5. Nicolás Castillo Triana (2024– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. *“Evaluation of speech characteristics and speech production network’s functioning in Parkinson’s disease patients treated with Magnetic Resonance-guided focused ultrasound subthalamotomy and pallidotomy”*. Supervisor
6. Gabriel Vásquez Viveros (2024– present), Ph.D. student, School of Engineering, Pontificia Universidad Católica de Chile. *“Automatic Doppler Ultrasound for clinical applications”*. Supervisor
7. Verónica Marín Bley (2023– present), Ph.D. student, School of Engineering, Pontificia Universidad Católica de Chile. *“Quantification of Pulmonary Aeration Using Ultrasound”* Supervisor
8. Laura Parra Núñez (2023 – present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *“Effects of anti-inflammatory therapies on the progression of coronary artery disease and its sex-dependent differences, in a murine model of diet-induced accelerated atherosclerosis”*. Supervisor.
9. Elizabeth Rivas Garrido (2023– present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Supervisor
10. Pablo Besa Vial (2022 – present), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *“Biomechanical assessment of internal fixation in tibial plateau fractures”*. Supervisor.
11. Pamela González Méndez (2022 – present), Ph.D. student, School of Medicine, Pontificia Universidad

- Católica de Chile. Thesis: *"Predictive MRI biomarkers of the effectiveness of high-intensity focal ultrasound (HIFU) subthalamotomy in patients with Parkinson's disease"*. Supervisor.
12. Juan Esteban Oyarzún Isamitt (2023 – 2026), Post-Doctoral Fellow, School of Medicine, Pontificia Universidad Católica de Chile. *"Therapeutic use of bee derivatives in the prevention of non-alcoholic fatty liver disease"*. Supervisor
 13. Cristian Montalba Zalaquett (2026), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. *"Predicting Working Memory Performance in Multiple Sclerosis with Diffusion-Weighted Imaging and Machine Learning"*. Supervisor
 14. Sebastián Pinto Ñanculeo (2026), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"MRI artifacts induced by titanium dental implants. An in vitro study at different magnetic fields"* Supervisor.
 15. Katherine Rivera Vega (2025), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Effect of diet and intestinal health on endotoxemia and its contribution to the development and progression of atherosclerosis"*. Supervisor.
 16. Pilar Vassiliu Hacke (2025), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Muscle MRS for the diagnosis of sarcopenia in elderlies"*. Supervisor.
 17. Elías Alfaro Nasta (2025), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Brain Volumetry as a predictive biomarker of the effectiveness of high-intensity focal ultrasound (HIFU) sub thalamotomy in patients with Parkinson's disease"*. Supervisor.
 18. Claudia Pazmiño Alarcón (2024), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Liver and muscle MRS at 3T for the diagnosis of Hypercortisolism"*. Supervisor.
 19. Laura Manjarrés Madrid (2024), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Sex differences in NAFLD progression in a metabolic model of liver steatosis"*. Supervisor.
 20. José Pensenti Anjel (2024), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Systemic inflammation in patient with chronic venous disease"*. Supervisor.
 21. Carlos Castillo Passi (2024), Ph.D. student, School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *"Diffusion Cardiac MR"*. Co-supervisor.
 22. María Belén Echenique (2023), M.Sc. student, School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *"A sound-based automatic system for cardiac and pulmonary rhythm analysis"*. Co-supervisor.
 23. Laura Parra Núñez (2023), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Sex differences in disease progression in a mice model of coronary disease and heart failure"*. Supervisor.
 24. Eduardo Godoy Sánchez (2023), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Clinical validation of a sound-based pulmonary Holter"*. Supervisor.
 25. Ignacio Rivera Silva (2023), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Decrease in work accidents and fatalities, and lower delays until first consult: the effect of COVID in one Chilean occupational health hospital"*. Supervisor.
 26. Héctor Ramos Zaldívar (2023), Ph.D. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *"Exosomes and nanoparticles in the study of brain lymphatic system"*. Supervisor.
 27. Leticia González Jara (2019–2021), Post-Doctoral Fellow, School of Medicine, Pontificia Universidad

Católica de Chile. *“Role of the sympathetic nervous system and a prooxidant environment in the development and clinical manifestations of coronary heart diseases”*. Supervisor

28. Katherine Rivera Vega (2020), M.Sc. student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *“Influence of the HDL SR-B1 receptor expression on the development of NAFLD in a murine model”*. Co-supervisor.
29. Aline Xavier Carvalho (2020), Ph.D. student, School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“Liver Fatty acids characterization in NAFLD, NASH and cirrhosis using MRS”*. Supervisor.
30. Gabriel Varela Mattatall (2019), Ph.D., School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“Accelerated Compressed Sensing in Diffusion Spectrum Imaging”*. Co-supervisor.
31. Carlos Milovic Fabregat (2018), Ph.D., School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“A noise-robust QSM reconstruction method using parallel MRI acquisitions”*. Co-supervisor.
32. Andrés Rodrigo G. (2018), undergraduate student, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *“Performance of dual-energy spectral mammography with contrast in the detection of breast cancer”*. Supervisor.
33. Luis Tapia R. (2018), Radiology resident, School of Medicine, Pontificia Universidad Católica de Chile. Thesis: *“Brain oedema quantification with CT”*. Supervisor.
34. Daniel Aguirre Reyes (2016), Ph.D., School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“Quantification of hepatic steatosis using Magnetic Resonance Imaging”*. Supervisor.
35. Alan Cid Olivares (2012), M.Sc., School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“Implementation of a Magnetic Resonance Elastography System”*. Co-supervisor.
36. Leonardo Ramírez Mac-Lean (2008), M.Sc., School of Engineering, Pontificia Universidad Católica de Chile. Thesis: *“A Technique for Reconstruction using Intensity Order (TRIO): An application to undersampled data”*. Co-supervisor.

Santiago, Chile, 2026